

Saving the Polar Bears: A Collaborative VR Storytelling Experience Enhanced by Multi-scale and Multi-view WiM Designs

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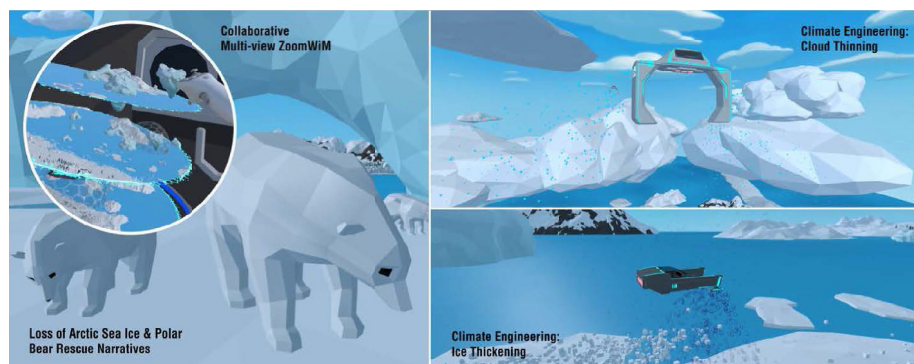


Figure 1: Illustration of our application advocating for environmental awareness of global warming and climate change.

ABSTRACT

The rapid loss of Arctic Sea ice has cascading impacts across environmental, ecological, and human systems, demonstrating the interconnected nature of global climate processes. Future climate engineering technologies have great potential to slow down the melting of Arctic Sea ice, yet few people are familiar with them. In this contest, we present a solution that utilizes the virtualness of VR experience to create a rapid cause and effect visualization in virtual environment (VE) for the application of the future climate engineering technologies. This educational experience features an engaging polar bear rescue storyline, gameplay mechanics, and two novel collaborative, adaptive zoom world-in-miniature 3D UIs that facilitate large-scale VE wayfinding, navigation, and task fulfillment.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Interaction techniques

1 INTRODUCTION

Virtual reality is a powerful tool to model rapid cause-and-effect scenarios for training and education. In the context of environmental sustainability education, this unique affordance is very useful in visualizing the effects of specific human activities on environmental changes, such as the use of fossil fuel and the greenhouse effect, which in the real world may take a long time to see the cause-and-effect relationship. Inspired by missions from a non-profit organization, Ocean Visions [1], which promotes innovation at the inter-

section of the ocean and climate crises, we are interested in visualizing how some of the potential future technologies Ocean Visions proposes can slow down the loss of Arctic Sea ice and mitigate its impact on local habitat for Arctic species in a collaborative and educational virtual environment. With that goal, and based on our prior success in using immersive VR storytelling and gamification to convey science concepts to the general public [2, 6], we created an enhanced WiM 3DUI with a multi-level of details zoom feature for interactions in a large-scale VE and a collaborative polar bear rescue mission to engage two users in using the tool. By combining the immersive and educational potential of VR with gamified storytelling and advanced multiplayer coordination, our system not only demonstrated technological innovations but also provided a compelling platform for collaborative learning about climate resilience.

2 NARRATIVES AND PROJECT SETUP

In a future where climate change has caused the Arctic ice to melt at an alarming rate, two brave adventurers (the players) find themselves on a quest to save the last remaining polar bears from an iceberg rapidly sinking into the sea. As the iceberg floods with rising sea water, they must work together to slow down the iceberg melting using climate engineering technologies such as cloud thinning, reflective aerosol injection, and ice thickening, to build a path for the polar bears to safely relocate before it's too late. The multi-user system uses Unity Netcode for GameObjects, Unity Relay, and Unity Lobby to synchronize avatars and interactions in a VR multiplayer environment. Each player's VR device tracking data (head and hands) maps to a Ready Player Me (half-body) avatar, enabling real-time updates for both the local user and other connected clients.

3 INTERACTION TECHNIQUES

The “World-in-Miniature” (WiM) 3D UI, proposed by Stoakley et al. [4], is an interaction technique in extended reality that allows users to interact with the virtual environment via a hand-held miniature replica. It enables dynamic viewport and intuitive interaction. A recent survey by Danyluk et al. on the design space of WiM suggested that although it is a widely-investigated 3D interface, there are still some gaps to be filled in both its design and application

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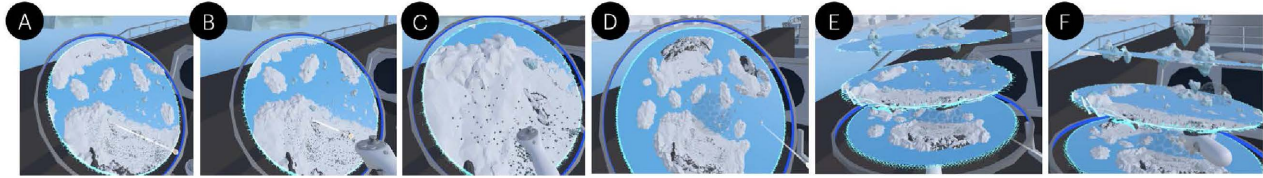


Figure 2: Illustration of the two techniques: (A-C) The *ZoomWiM* technique, in which the WiM adaptively zooms in to the ray intersection point when approached by the controller; (D-F) the *MultiWiM* technique in which user could mark an area to create another WiM instance stacked on top with the selected view. All instances are synchronized and could be interacted.

domain. First, there has been a lack of WiM designs that support interacting with large-scale VEs on different levels of granularity. For example, a user may reference the entire replica of the VE for awareness of other users' locations, and zoom in to manipulate the placement of an object. Second, few existing research explored applications of WiMs in collaborative multi-user setups. In collaborative experiences, one player often needs various levels of awareness of other player's locations and actions, which traditional static WiM may fall short. In this work, we proposed two enhancements to the traditional WiM 3D UI with adaptive multi-scale and multi-view enhancements to fill these gaps.

3.1 ZoomWiM technique

To facilitate interaction with large-scale VEs at varying levels of granularity using a WiM interface, we introduce *ZoomWiM*, an adaptive, scrollable, and zoomable WiM technique. Unlike earlier work by Wingrave et al. [5], which relied on manual gestures for scrolling and scaling, our approach dynamically adjusts the WiM's level of detail (LOD) and focus. The WiM, which is located on the user's non-dominant hand, initiates with a fully zoomed-out miniature of the global VE. When the user brings their dominant hand closer to the WiM (shorter than 30cm), it automatically zooms in, centering on where the ray from the dominant-hand controller intersects the WiM. The zoom scale linearly correlates with the distance between the dominant and non-dominant hands or controllers (please see Figure 2 (A-C)). The zooming in stops when two hands distance is below 5cm, so the user could perform precise manipulations on the individual objects within the large VE. As such, *ZoomWiM* intuitively maps actions involved in direct manipulation techniques to modifications on the WiM viewport, enhancing intuitiveness and reducing the cognitive and physical effort required for back-and-forth global and local context switching.

3.2 MultiWiM technique

Another adaptation of WiM took inspiration from [3]. Rather than relying on scrolling and zooming within a single-view WiM, we introduce *MultiWiM*, a technique that utilize multiple WiM views to reveal details at various locations and scales. In *MultiWiM*, users can employ the controller in their dominant hand to mark a region on the WiM located on their non-dominant hand, triggering the creation of new WiM instances that zooms into the selected area (please see Figure 2 (D-E)). These WiMs are synchronized, sharing a common reference to the original VE, ensuring that any changes made in one WiM are seamlessly reflected across all multi-view instances (please see Figure 2 (F)). This allows users to manipulate small objects in a fully zoomed-in WiM while simultaneously being aware of their collaborators in a zoomed-out WiM for global context. Additionally, objects can be moved across WiMs for rapid interactions. By enabling concurrent access to multiple WiMs, *MultiWiM* augments users with additional layers of awareness, facilitating large scene exploration, collaborator awareness, and fine-grained interactions with individual small objects.

4 MULTIPLAYER COLLABORATIVE STORYTELLING DESIGN

The multiplayer cooperation aspect of the system was designed to highlight teamwork and strategic coordination. By leveraging the

enhanced WiM interfaces within the virtual environment, players engaged in collaborative tasks utilizing adaptive zoom interactions and multi-level selection zoom. These features enabled players to adopt distinct roles and responsibilities, fostering a dynamic and engaging multiplayer experience.

Designing effective collaborative VR storytelling tasks for science learning represents a key focus of this project. Unlike single-user storytelling experiences, where NPC facilitation can seamlessly integrate learning tasks into the narrative, collaborative VR experiences require careful design to engage users working in different virtual locations. Each user must remain invested in their assigned roles and tasks while contributing toward a shared goal. To address this challenge, we developed two tasks that allow users to alternate between the roles of planner and field operator. In Task 1, the planner role user uses the WiM to set waypoints that guide the field operator role user to locate polar bears. In Task 2, the roles reverse: the former field operator user becomes the planner user, signaling from the polar bear location to guide the previous planner user—now the field operator—to designated aerial space to perform a cloud thinning task.

5 CONCLUSION

To sum up, by combining the advanced adaptive multi-scale and multi-view WiM interfaces to enhance multiplayer cooperation mechanics and a compelling narrative, our application will effectively engage users in understanding and addressing the critical issue of climate change. Our work demonstrates the advantages of using VR as a medium to foster environmental awareness and collaboration, paving the way for future applications in education and beyond.

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